



Applications of VaR

Risk Management in Financial Institutions

Lecture 4_2

Béla ÖCSI

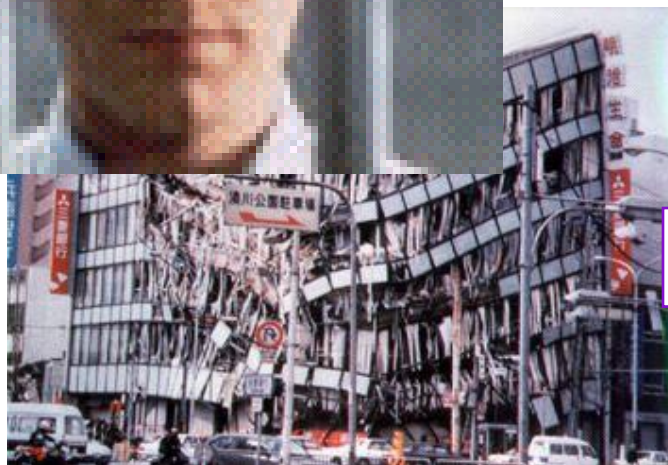
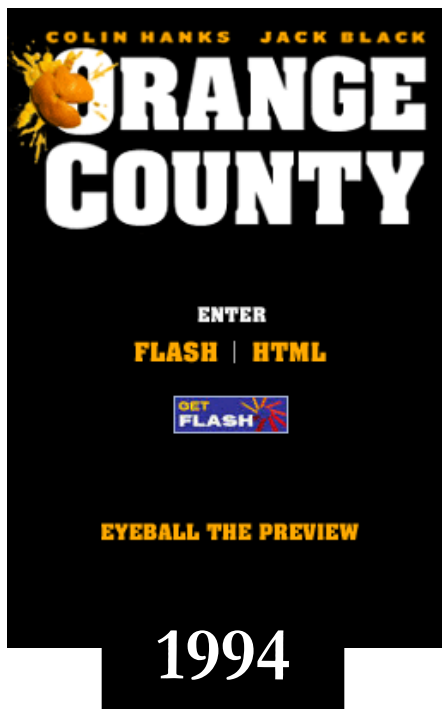
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What has drawn attention to the risks



1993

2001 dec



JPMorgan and the VAR world

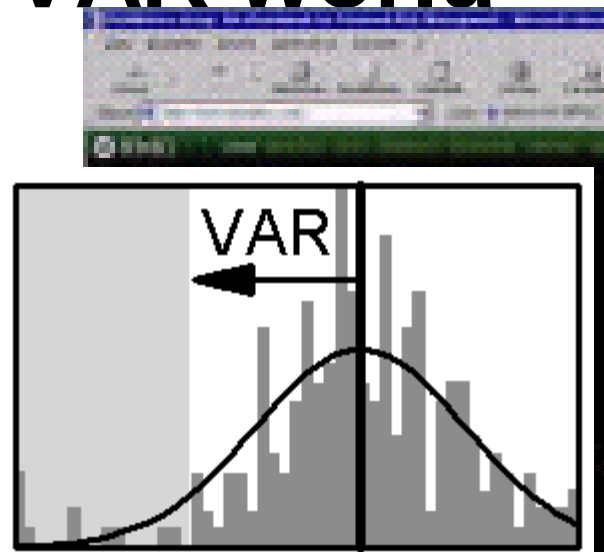


John Pierpont (J.P.)
Morgan (1837-1913)



Sir Dennis
Weatherstone -
1994

4.15 TM



What can the Value at Risk be applied for?

- Margin requirements
- Capital allocation on VaR basis
- Limit structure (V@R limits)
- Return/risk analysis (RORAC)
- Backtesting
- Risk of nontrading portfolios (ERM, CFaR)

Margin requirements

- For derivatives
- To hedge the effects of price changes
- VaR-based methodologies

*„What is the initial margin requirement?
To open 1 contract BUX futures
position, the margin required is twice
the current margin requirement set by
Keler Zrt.”*



A SPAN számítás paramétereit a KELER KSZF a leirataiban teszi közzé.

VaR számítás:

- Visszatekintési időszak: legalább 1 év (250 nap), mely stressz időszakot tartalmaz
- Konfidencia szint: 99%
- Likvidációs periódus: 2 nap
- Prociklikussági puffer: 25%
- Tolerancia szint: 1%
- Igazodási faktor: 98,17% (250 napos visszatekintési periódus esetén)

1.4. Alapbiztosíték mértékének meghatározása

Biztosítékok meghatározása során megkülönböztetünk vezető termékeket, illetve nem vezető és illikvid termékeket. A biztosítékszámítási módszertanban alkalmazott jelölések:

1. $KSzFmargin_t$: a szakértői és az (il)likviditási pufferrel növelt VaR érték
2. $PROmargin_t$: a 25%-os prociklikussági pufferrel növelt $KSzFmargin_t$
3. $margin_t^{pro-felold}$: biztosíték értéke a prociklikussági puffer elengedése esetén
4. $margin_t^{pro-visszaepitese}$: biztosíték értéke a prociklikussági puffer visszaépítése során
5. $MINmargin_t$: a biztosíték mindenkor minimális értéke
6. $MAXmargin_t$: a biztosíték mindenkor maximális értéke
7. $margin_t$: az adott napra érvényes tényleges biztosíték



Capital allocation

○ Risk-based capital (requirement) (economic capital):

- Determined by banks using their own methods
- calculated in order to cover losses on different positions, arising from different risks
- Basis for internal limits

○ Regulatory capital (requirement):

- Trading book (2013/575 EU regulation (CRR) (*THE EUROPEAN PARLIAMENT'S AND THE COUNCIL'S 575/2013/EU REGULATION (2013. June 26.) on prudential requirements for credit institutions and investment firms and amending the 648/2012/EU regulation*)
 - standard model (simplified regulatory "economic capital model")
 - internal model (fixed parameter own economic capital model)
- FRTB

Limits: a major bank's VaR&Limit Report

Trader Report

Position date: 20 | Portfolio: CEE_HU_Limit Currency: EUR

Name: CEE_HU_Limit

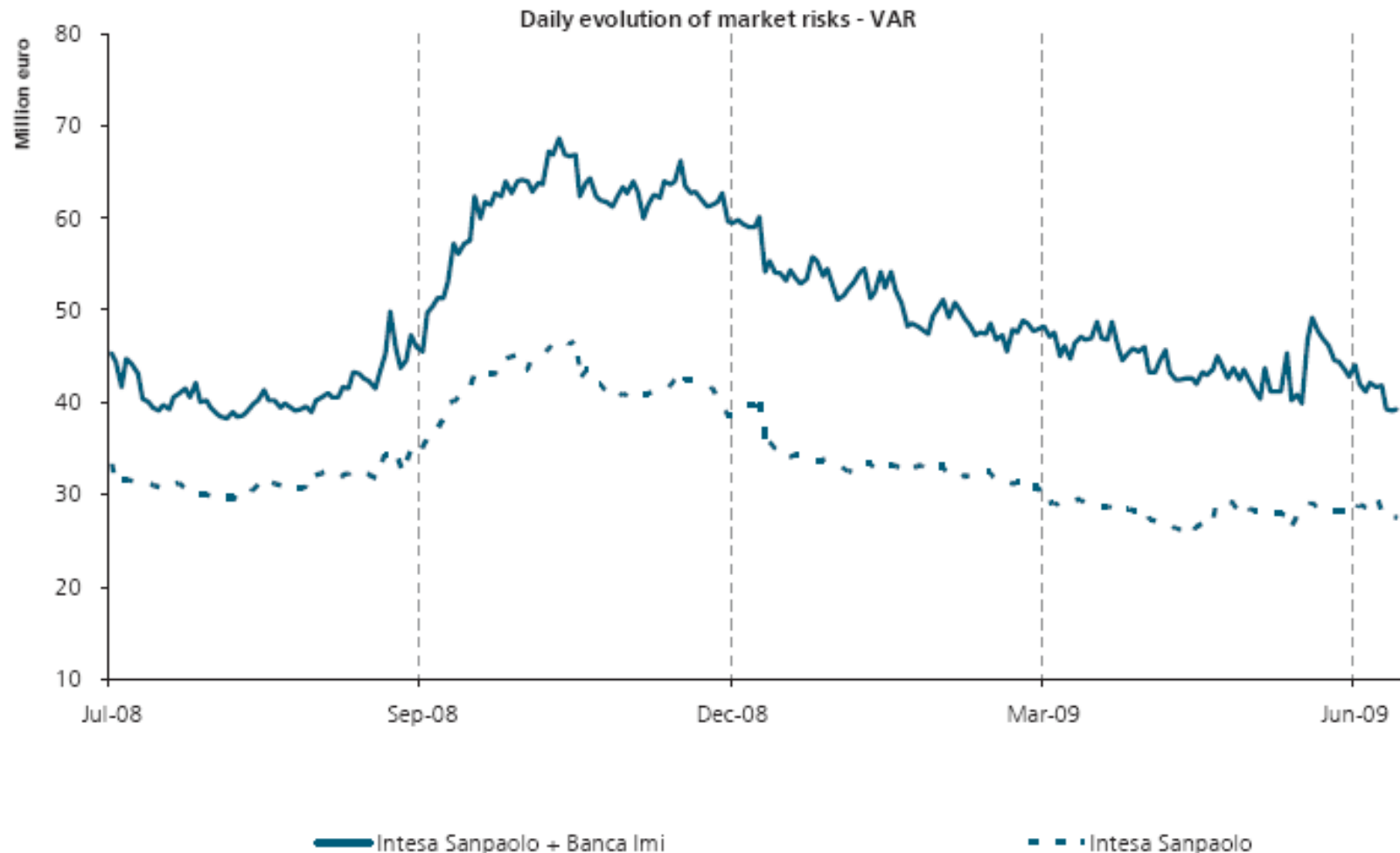
Simulation Summary

Type	Avg	2.33 Std	99% Perc	Error	Avg ov 99%	Skew	Kurtosis
FXIRSPEQ +SMR +QT incl. ResVar (SP, EQ)	5,856	983,940	-927,891	18,964	-973,999	0.17	-0.05
FX	-3,829	116,834	-125,831	12,911	-154,916	-0.02	0.70
IR excl. Spreads	18,159	250,640	-252,294	35,220	-393,686	-0.08	21.50
SP incl. ResVar	-81,694	958,520	-982,821	34,769	-1,063,783	0.18	-0.01
Vega Internal Model	3	222	-223	35	-312	0.28	3.43
SP excl. ResVar	-69,369	172,398	-332,953	46,511	-410,262	-0.32	7.48
SP ResVar only	-12,373	951,962	-878,719	48,290	-977,988	0.21	0.03
FXIR excl. Spreads	18,215	240,431	-216,308	13,224	-332,749	-0.07	23.28
FXIRSPEQ excl. ResVar (SP, EQ)	18,312	292,655	-286,621	76,019	-454,945	-0.11	11.44
FXIRSPEQ +SMR +QT incl. ResVar (SP)	5,856	983,940	-927,891	18,964	-973,999	0.17	-0.05
CAD Vega	23	655	-515	24	-538	0.74	0.96
CAD All	-24,585	3,104,259	-2,972,554	156,334	-3,248,994	0.16	-0.01

Type	Value
Limit VaR	-928,113.74
Limit	2,200,000.00
Limit usage	42.19
Max. FX Volume Limit Usage	43.17
Max. EQ Volume Limit Usage	
Max. BPV Volume Limit Usage	47.18

VaR in practice, in a major bank, during the crisis

Explanatory notes – Risk management



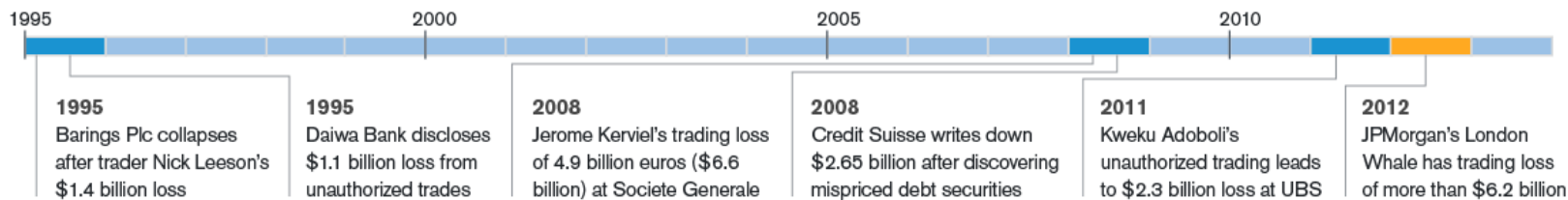
Source: IntesaSanpaolo Half-year report, 30th June 2009

The London Whale, or VaR models in real life



„The trader known as the London Whale lost at least \$6.2 billion for JPMorgan Chase & Co. in 2012. What Bruno Iksil and his colleagues did was the same old story — **doubling down after a loss with bigger and bigger bets**. But plenty more was wrong. They worked in a part of the bank, the Chief Investment Office (CIO), whose job was to hold down the bank's risk level. Instead, the CIO used the \$350 billion it had to invest to become a moneymaker, with its London office focused on **complex credit derivative trades** that had less and less to do with hedging. The trouble came in early 2012, when the bank decided to **reduce the risk** in the London swaps portfolio by making more offsetting bets. As the strategy unraveled, Iksil's positions grew so big that they **disrupted the thinly traded markets** he worked in — earning him nicknames of Whale and Voldemort, and making his group's hard-to-unwind trades a **target for hedge funds**. After the trades collapsed, regulators found that Iksil's colleagues had been keeping two sets of books to minimize the projected size of the losses.”

Biggest Bank Trading Losses



What the VaR limits showed...

Value-at-Risk for the CIO (10Q VaR)

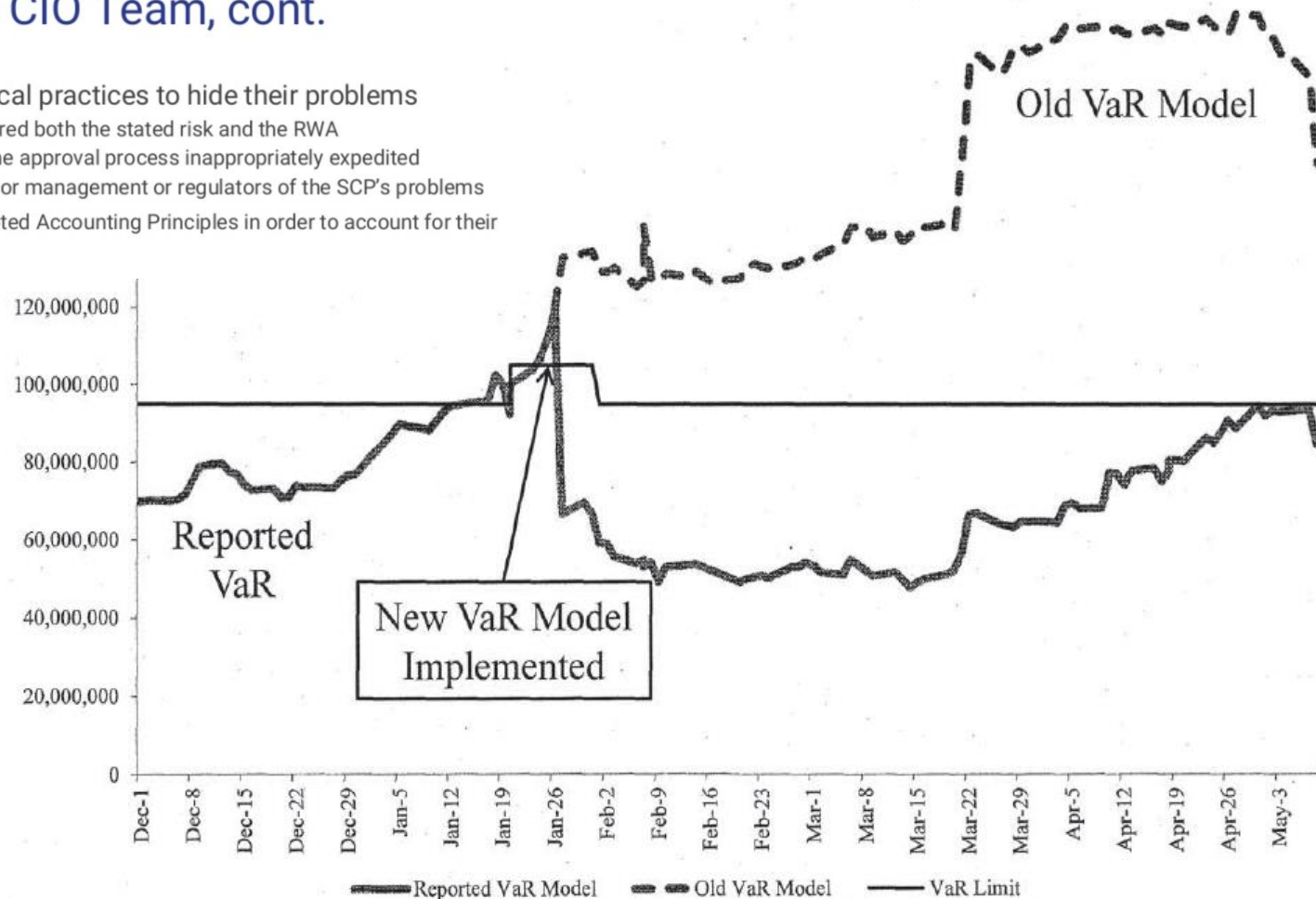
on Whale Presentation

Mistakes Made by the CIO Team, cont.

- The CIO team engaged in unethical practices to hide their problems
 - Created a new VaR model that lowered both the stated risk and the RWA
 - Successfully fought to have the approval process inappropriately expedited
 - Did not adequately inform JPM senior management or regulators of the SCP's problems
 - Departed from U.S. Generally Accepted Accounting Principles in order to account for their derivatives more favorably

“(VaR) and other **risk limits**, for instance, **were breached** more than 300 times before the bank switched to a **new vaR model**, to a more lenient risk-evaluation formula — one that **underestimated risk by half because of a spreadsheet error.**”

Permanent Subcommittee on Investigations
EXHIBIT #1e



Source: Subcommittee chart created from data provided by JPMorgan Chase, JPMC-Senate/Levin 000155-6.

RORAC: Analysis of the Return On Risk-Adjusted Capital

Asset manager „A”

- HUF 2 Bn portfolio, annual profit of HUF 260 M, average V@R (99%, 10 days) HUF 320 M
- ROA = 13,0 %

RORAC: 81,3 %

(Return on Risk-Adjusted Capital)

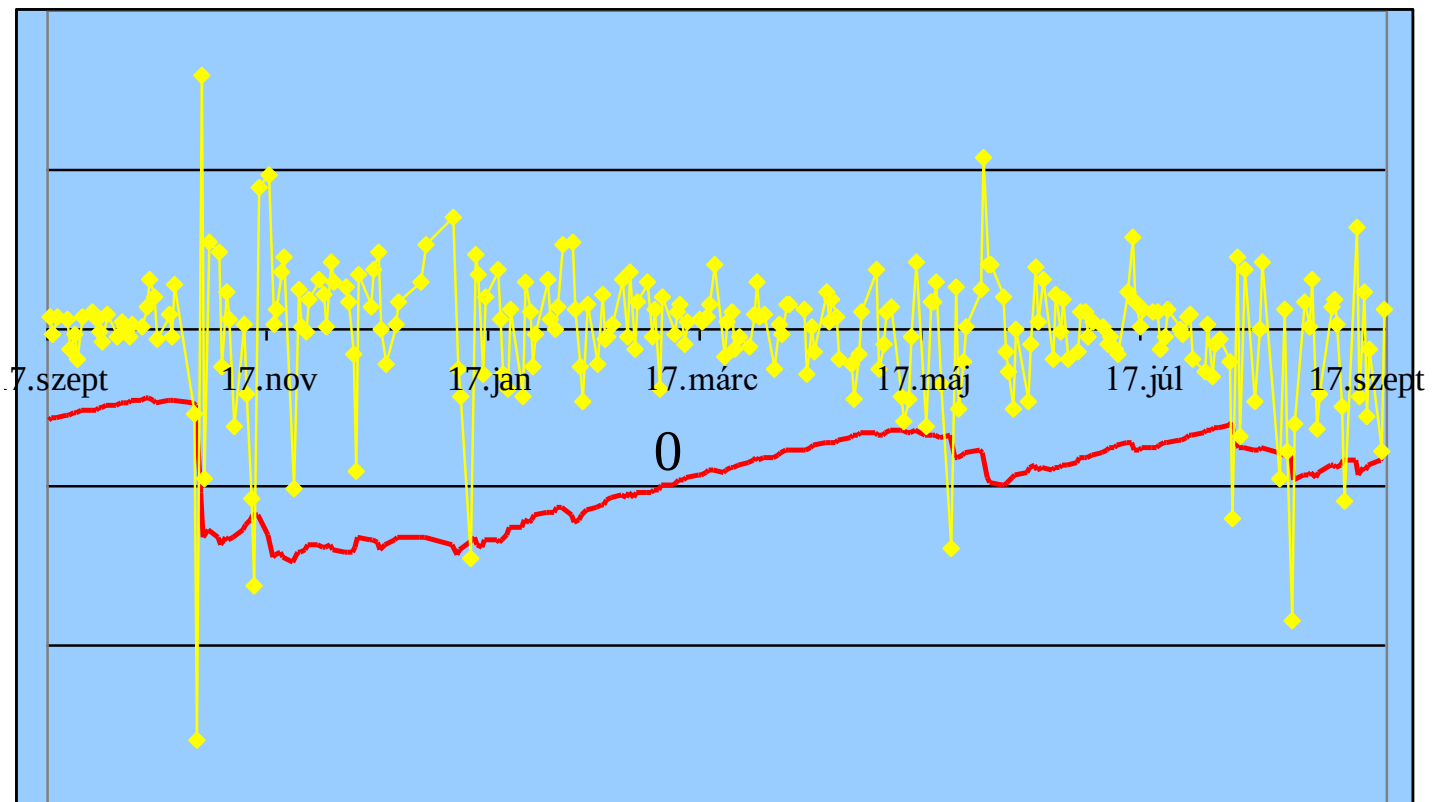
Asset manager „B”

- HUF 2 Bn portfolio, annual profit of HUF 200 M, average V@R (99%, 10 days) HUF 225 M
- ROA: 10 %

RORAC: 88,9 %

Model verification - back testing

Analysis of the number of VAR violations

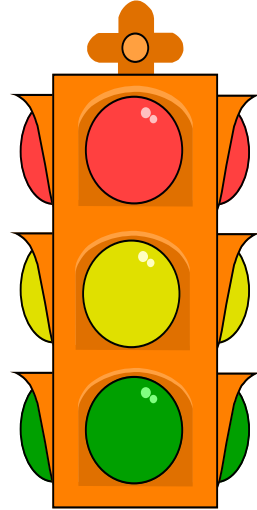


Retrospective testing - back testing

- *Test appropriate*: accepts good model, rejects bad model
- *First type error*: rejects the good model
(*measured probability of first type error is about 11%*)
- *Second type error*: accepts the bad model
(*measured probability of second type error 15-35%*)

Model verification and regulation

- Every quarter, the 1-day 99% VAR is compared with the actual P/L of the previous 250 trading days
- zones (out of 250 days) - if there are too many violations, the k value used for regulatory capital is increased
 - 0-4 violations: $k=3$ (*green*)
 - 5-9 violations: k increases to 4 (*yellow*)
 - Over 10 violations: model recalibration (*red*) (*Hungary only at 25*)



ERM Definition

“Enterprise risk management is a process, effected by an entity’s board of directors, management, and other personnel, applied in strategy setting and across the enterprise, designed to identify potential events that may affect the entity, and manage risk to be within its risk appetite, to provide reasonable assurance regarding the achievement of entity objectives.”

For a Fund Manager...(equation 27.1, page 565)

- Key risk appetite question could be: What is the return, R , that we want to be exceeded with a high probability p
- If R_M is the return from the market, R_F is the risk-free return, and σ_M is the standard deviation of the return from the market, then the β of the portfolio should be

$$\frac{R - R_F}{R_M - R_F + N^{-1}(1 - p)\sigma_M}$$



**Thank you for your
attention!**

